1 AI Generated

GPT 6.1 Sol High Effort

`https://chatgpt.com/g/g-p-6aab0741ce3c8191b7ba47a512862502-machine-learning/c/6aedb7fd-3ea8-83e8-9d12-58bc3f930671`

2 Prompt

Use the attached latex as the basis for a neural network diagram but make the following changes:

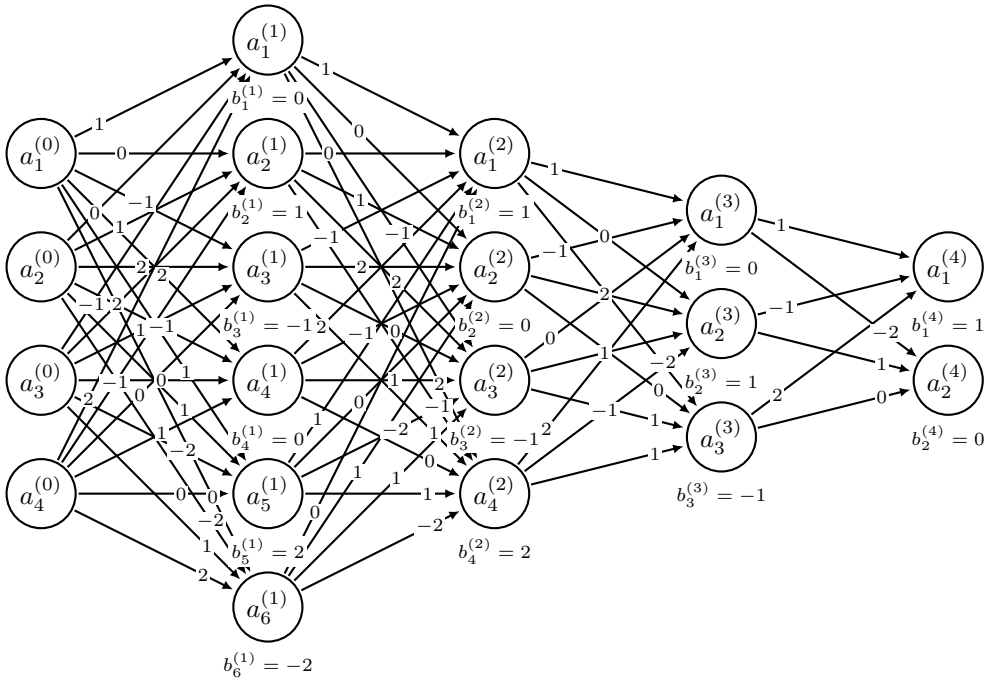
Label every layer with ℓ like I started attempting

Define and dynamically lookup the weights that should be used instead of 100.

Explain the comments I had enumerated I didn't understand.

Change all of the colors to regular black and white text, but annotate what would need to change in order to restore the colors with inline comments

Input layer	Hidden layer	Hidden layer	Hidden layer	Output layer
$\ell = 0$	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$
$\mathbf{a}^{(0)} = \mathbf{x}$	$\phi^{(1)} = \text{ReLU}$	$\phi^{(2)} = \text{ReLU}$	$\phi^{(3)} = \tanh$	$\phi^{(4)} = \sigma$



$$z_i^{(\ell)} = \sum_{j=1}^{n_{\ell-1}} w_{ij}^{(\ell)} a_j^{(\ell-1)} + b_i^{(\ell)}, \qquad a_i^{(\ell)} = \phi^{(\ell)}\left(z_i^{(\ell)}\right)$$

$$\mathbf{z}^{(\ell)} = W^{(\ell)}\mathbf{a}^{(\ell-1)} + \mathbf{b}^{(\ell)}, \qquad \mathbf{a}^{(\ell)} = \phi^{(\ell)}\left(\mathbf{z}^{(\ell)}\right) \quad \text{(elementwise)}$$

$$\text{ReLU}(u) = \max\{0, u\}, \qquad \tanh(u) = \frac{e^u - e^{-u}}{e^u + e^{-u}}, \qquad \sigma(u) = \frac{1}{1 + e^{-u}}$$